Unit 3 · Outcome 2 · Sustainable Development

Sustainability principles, applied to real management

KK03 VCE Environmental Science · Units 3 & 4 · Exam study summary

Key knowledge. · dot point 3 of 8 00 What this dot point actually asks The study design lists this dot point as: sustainability principles as they apply to environmental management — conservation of biodiversity and ecological integrity; efficiency of resource use; intergenerational equity; intragenerational equity; the precautionary principle; and the user-pays principle

Digital Vector EnviroSci · single-topic mastery summary — clean explanations, comparison tables, visual aids, exam traps, command words and worked answers.

KEY KNOWLEDGE · DOT POINT 3 OF 8

00 What this dot point actually asks

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Notice the three words that decide your marks: as they apply. VCAA isn't asking you to recite six definitions — a glossary does that. They want you to take a real management decision and show how each principle either supports it, challenges it, or is traded off against another. Definition earns the first mark; application to a case study earns the rest.

> ◆ **KEY INSIGHT**
>
> The shape of the dot point Principle ➔ definition ➔ applied to a named case study ➔ judged (upheld or compromised?). If your answer stops at the definition, you've stopped at one mark.

These same six principles reappear in biodiversity conservation (U3·O1), energy resources (U4·O2) and the final effectiveness dot point of this very chapter (KK08). Lock them down here once, and you've banked marks across four outcomes.

.56 T CO_2 / MWH

01 The running case study — Latrobe Valley after coal

To keep every principle concrete, this whole chapter uses one decision you can picture: Victoria is shutting its brown-coal power stations, and now has to decide how to manage the land, the workers and the emissions left behind.

The Latrobe Valley, about 135 km east of Melbourne, sat on one of the world's largest brown-coal (lignite) deposits and powered Victoria for a century. Hazelwood Power Station — eight 200 MW units, up to a quarter of Victoria's base-load electricity — was rated the least carbon-efficient power station in the OECD, releasing roughly 1.56 t CO_2 / MWh and about 16 Mt CO_2 per year, around 14% of Victoria's emissions.

Hazelwood closed in 2017. Yallourn is due to close in 2028 and Loy Yang around 2035. That leaves three management problems that map almost one-to-one onto the six principles:

> ◆ **NOTE**
>
> Why this case study earns marks It's real, current and Victorian, so examiners accept the detail. Better still, it forces trade-offs — almost every principle pulls against another here, which is exactly where the "analyse" and "evaluate" marks live.

$439 M

02 The six principles — definition and application

Learn each one as a pair: the VCAA definition (the wording examiners reward) and the Latrobe Valley application (the marks beyond the first). The card colours match the 3D lens in the next section.

03 Interactive · the principle lens

P · POINT answer the verb → **E · EVIDENCE** example / mechanism → **L · LINK** tie back to question

The P·E·L answer shape — make a Point, support with Evidence, then Link to the question.

> ✎ **P·E·L WRITING**
>
> Each principle is a different lens on the same decision. Click a lens to see how that principle "reads" the Latrobe Valley transition — and where it conflicts with the others.

> ◈ **ANALOGY**
>
> Analogy · six referees, one match Imagine the same football match watched by six referees, each only allowed to enforce one rule. The "efficiency" ref loves the renewables plan; the "intragenerational equity" ref blows the whistle if Morwell workers are abandoned; the "user-pays" ref checks who pays for the mess. A sustainable decision is one that keeps all six reasonably satisfied — not one that wins on a single rule.

04 Interactive · the equity trap

> ⚠ **EXAM TRAPS**
>
> The single most common mistake on this dot point is swapping inter generational and intra generational equity. They sound identical and they're worth different marks. This scene separates them on two axes.

> ◆ **KEY INSIGHT**
>
> The memory hook that ends the confusion INTER = bet ween generations → fairness along the time axis (us → 2050 → 2100). INTRA = inside today's generation → fairness across people and places right now (Morwell vs Melbourne vs developing nations). Intra is inside the present.

05 Applying the principles to management

Real management can't maximise all six at once — they pull against each other. "Applying" the principles means naming the trade-offs and judging the balance, which is exactly what "analyse" and "evaluate" questions reward.

Three tensions run through the Latrobe Valley decision:

> ◆ **NOTE**
>
> Tension 1 · ecological integrity vs intragenerational equity Closing coal fast is best for the climate and the void, but it can strand Morwell's workers. Managing well means pairing the environmental goal with the Latrobe Valley Authority's retraining — you uphold both P1 and P4 rather than sacrificing one.

◆ **NOTE**

Tension 2 · user-pays vs intergenerational equity If the rehabilitation bond ($73 M) is far below the real cost ($439 M), the polluter under-pays now and future taxpayers inherit the bill — a failure of P6 that becomes a failure of P3. Victoria's 2026 trailing-liabilities law tries to keep the cost on the operator, not the grandchildren.

◆ **NOTE**

Tension 3 · precaution vs efficiency Flooding the void into a pit lake is efficient (it stabilises the land cheaply) but uncertain (huge water draw from the Latrobe River). The precautionary principle says: run the Environment Effects Statement first, even though it's slower — serious, possibly irreversible harm outweighs the delay.

✓ **TIP**

Exam move: when a question says "evaluate the management against the sustainability principles", walk through them, but spend your words on the two or three that conflict. A judgement that names a trade-off beats a list that names all six.

06 Practise · spot the principle

Each card describes a real management action. Decide which principle it most clearly serves. Instant feedback explains the trap.

07 Command-word decoder

◈ **COMMAND WORDS**

The verb in a question sets the ceiling on your marks. Tap each to see what the examiner is really asking for on this dot point.

Command word	What the marker wants
Identify / State	Name it — no explanation.
Describe	Give the features / what happens, in order.
Explain	Give the how / why — cause and effect.
Distinguish / Compare	State the difference (and similarity) explicitly.
Analyse	Break into parts and show how they relate.
Evaluate / Justify	Weigh strengths & limits, then a reasoned verdict.

Match your answer to the verb — the fastest marks in the exam.

08 Exam traps to avoid

⚠ EXAM TRAPS

The number-one error. Fix: inter = across time (future generations); intra = across people now (same generation, including poorer nations). It means the opposite. Fix: uncertainty is not a reason to delay action against serious or irreversible harm — act early. It's about who bears the environmental cost. Fix: the polluter/user pays for repairing impacts (e.g. rehabilitation bonds), so the cost isn't shifted to the public. Fix: it's minimising the resources, energy and waste per unit of benefit — an environmental measure, not only a financial one. Fix: integrity is the health and functioning of the whole ecosystem — its processes and services — not just the count of species. A bare list caps you at definition marks. Fix: tie every principle to a named case-study action and say whether it's upheld or compromised.

09 P·E·L — a sentence shape for full marks

The P·E·L answer shape — make a Point, support with Evidence, then Link to the question.

✎ P·E·L WRITING

For any "apply / analyse / evaluate the principles" question, each principle you discuss should move through three beats. One principle, three sentences, three marks. State which principle and whether the management upholds or compromises it. Quote a number, body or action from the case study. This is where most marks are won or lost. Explain what that means for sustainability, or who ends up bearing the cost.

✓ TIP

Why it works: the Point earns the definition mark, the Evidence earns the application mark, and the Link earns the analysis/evaluation mark — the three marks examiners separate in their reports.

10 Same question, two answers

Question (4 marks): "Brown-coal generation in the Latrobe Valley is being replaced with renewables. Analyse how two sustainability principles support this decision."

Replacing coal with renewables is good for sustainability. It helps efficiency of resource use because renewables are more efficient. It is also fair for everyone and good for the environment, so it follows the sustainability principles and should be done.

● NOTE

Why it loses marks: only one principle is genuinely identified; "more efficient" and "fair for everyone" are asserted with no case-study evidence and no analysis. "Good for the environment" is vague. This is a list, not an analysis.

Efficiency of resource use is improved: Hazelwood was the OECD's least carbon-efficient station at ~1.56 t CO_2/MWh, so replacing it with wind and solar delivers more useful energy per unit of emissions and water.

Intergenerational equity is also supported: cutting roughly 16 Mt CO_2 a year reduces the climate burden passed to future generations, helping them meet their own needs — the core of that principle.

> ● **NOTE**
>
> Why it earns full marks: two principles correctly named and defined, each tied to a specific figure from the case study, each linked to why it advances sustainability. Two principles × (point + applied evidence) = 4.

11 Exam-style questions

> ★ **PRACTICE & SELF-MARK**
>
> Five graded questions, 19 marks total. Attempt each in your logbook first, then reveal the model answer, marking guide, and self-mark. Your score follows you in the dock.

EXAM TECHNIQUE

★ Worked answers — weak vs strong

The same question answered two ways. The strong column shows the moves that earn the marks.

P · POINT answer the verb → **E · EVIDENCE** example / mechanism → **L · LINK** tie back to question

The P·E·L answer shape — make a Point, support with Evidence, then Link to the question.

✗ WEAK ANSWER

Replacing coal with renewables is good for sustainability. It helps efficiency of resource use because renewables are more efficient. It is also fair for everyone and good for the environment, so it follows the sustainability principles and should be done. Why it loses marks: only one principle is genuinely identified; "more efficient" and "fair for everyone" are asserted with no case-study evidence and no analysis. "Good for the environment" is vague. This is a list, not an analysis.

✓ STRONG / MODEL ANSWER

Efficiency of resource use is improved: Hazelwood was the OECD's least carbon-efficient station at ~1.56 t CO_2/MWh, so replacing it with wind and solar delivers more useful energy per unit of emissions and water. Intergenerational equity is also supported: cutting roughly 16 Mt CO_2 a year reduces the climate burden passed to future generations, helping them meet their own needs — the core of that principle. Why it earns full marks: two principles correctly named and defined, each tied to a specific figure from the case study, each linked to why it advances sustainability. Two principles × (point + applied evidence) = 4.

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Worked good/weak answers illustrate exam technique — always check against current VCAA marking advice.